Ejercicios Certamen 3 Mecánica Clásica

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June 22, 2026

1 Problema 1

The transformation of coordinates and momenta has a form

$$P = \alpha \frac{p^2 q}{\sqrt{1+p^2q^2}}, \qquad Q = \beta \frac{\sqrt{1+p^2q^2}}{p}, \qquad \alpha, \beta = \text{const} \tag{1.1}$$

a. Find the values of constants α, β for which the transformation is canonical. For these values of α, β, find explicitly the generating function F which corresponds to the transformation.
Respuesta: Podemos calcular el corchete de Poisson entre $[P, Q]_{p,q}$, es cual deberá ser cero. Usando `Mathematica` tenemos

$$[P, Q]_{p,q} = \alpha\beta \tag{1.2}$$

Por lo tanto, para que la transformación sea canónica deberá cumplirse que

$$\beta = \frac{1}{\alpha} \tag{1.3}$$

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b. Apply the transformation found in the previous item to the hamiltonian of harmonic oscillator $H(p,q) = \frac{p^2}{2m} + \frac{kq^2}{2}$ and find the hamiltonian $\tilde{H}(P,Q)$ in terms of the new variables (P,Q). Write out the canonical equations in terms of these new variables.
Respuesta:

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